

Xi Ursae Majoris
1.5 Suns

Megrez
25 Suns

Merak
60 Suns

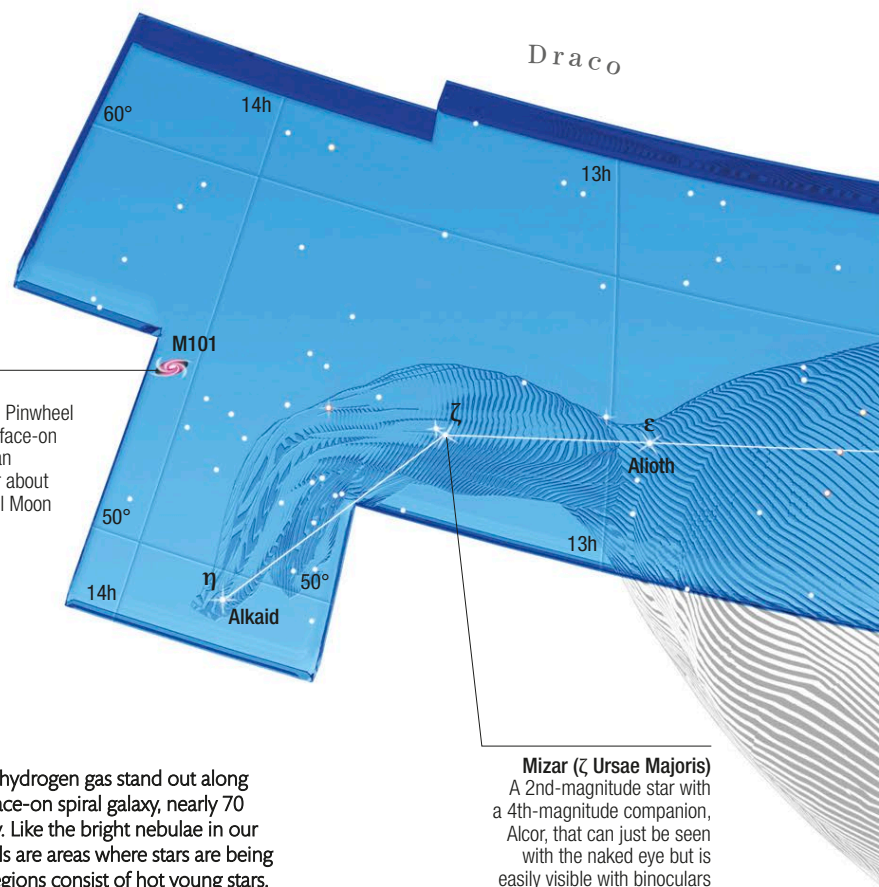
Phad
62 Suns

URSA MAJOR THE GREAT BEAR

THE THIRD-LARGEST CONSTELLATION, URSA MAJOR IS BEST KNOWN FOR CONTAINING THE BIG DIPPER (ALSO CALLED THE PLOUGH), PROBABLY THE MOST FAMOUS STAR PATTERN IN THE ENTIRE SKY.

Seven stars make up the familiar ladle-shaped pattern known as the Big Dipper: Dubhe, Merak, Phad, Megrez, Alioth, Mizar, and Alkaid. The second star in the handle of the dipper is a wide double. The brighter of the pair is Mizar, and its companion is Alcor. Two stars in the bowl of the dipper, Merak and Dubhe, point toward the north pole star, Polaris, in nearby Ursa Minor, the Little Bear.

Ursa Major also contains several interesting deep-sky objects. These include M101, a face-on spiral also known as the Pinwheel Galaxy; M81 and M82 (also called the Cigar Galaxy), a pair of galaxies that are thought to have had a close encounter about 300 million years ago; and the planetary nebula M97, popularly called the Owl Nebula because of its resemblance to an owl's face.

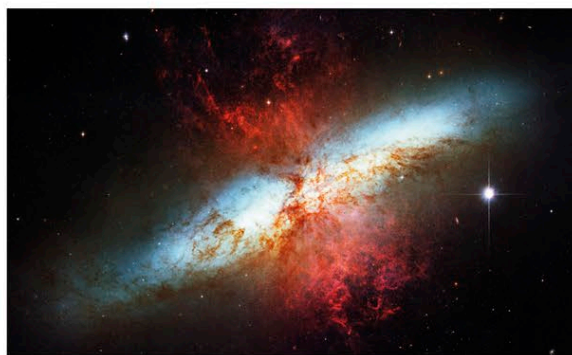
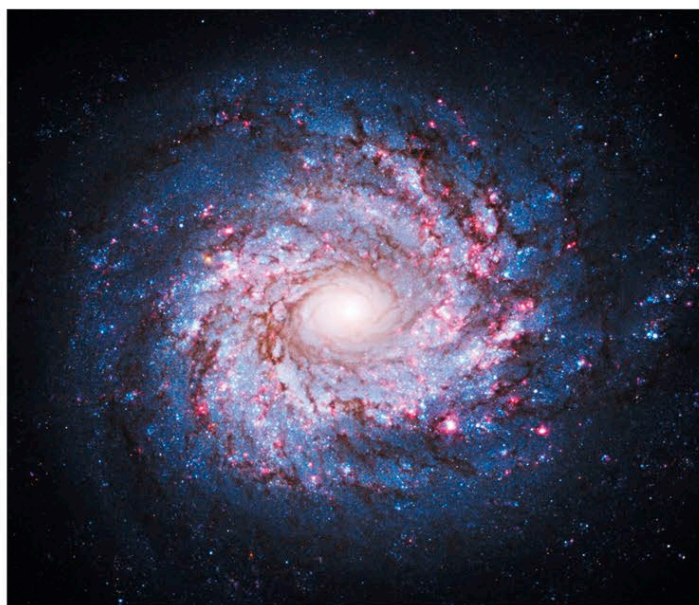


M101
Also known as the Pinwheel Galaxy, this large, face-on spiral galaxy has an apparent diameter about half that of the Full Moon

◀ NGC 3982

Pink clouds of glowing hydrogen gas stand out along the spiral arms of this face-on spiral galaxy, nearly 70 million light-years away. Like the bright nebulae in our own galaxy, these clouds are areas where stars are being born, while the bluer regions consist of hot young stars. NGC 3982 is about 30,000 light-years wide, nearly one-third of the diameter of the Milky Way.

Mizar (ζ Ursae Majoris)
A 2nd-magnitude star with a 4th-magnitude companion, Alcor, that can just be seen with the naked eye but is easily visible with binoculars



◀ M82

Popularly known as the Cigar Galaxy, this is undergoing a huge surge in star formation as a result of an interaction with its neighboring galaxy, M81. Plumes of hot, ionized gas (red in this Hubble image) are being blasted out above and below the disk of the Cigar Galaxy. Situated in the northern part of Ursa Major, both galaxies are 12 million light-years from Earth.

▷ Star distances

Ursa Major's main pattern stars lie between 29 and 358 light-years away from Earth. The two stars that form the ends of the Plough asterism—Dubhe and Alkaid—are 123 and 104 light-years away, respectively. The other five stars of the asterism—Merak, Phad, Megrez, Alioth, and Mizar—all lie at similar distances (about 80–86 light-years away) and are moving in the same direction through space. They form what is known as the Ursa Major Moving Group, a former open cluster that has drifted apart.

